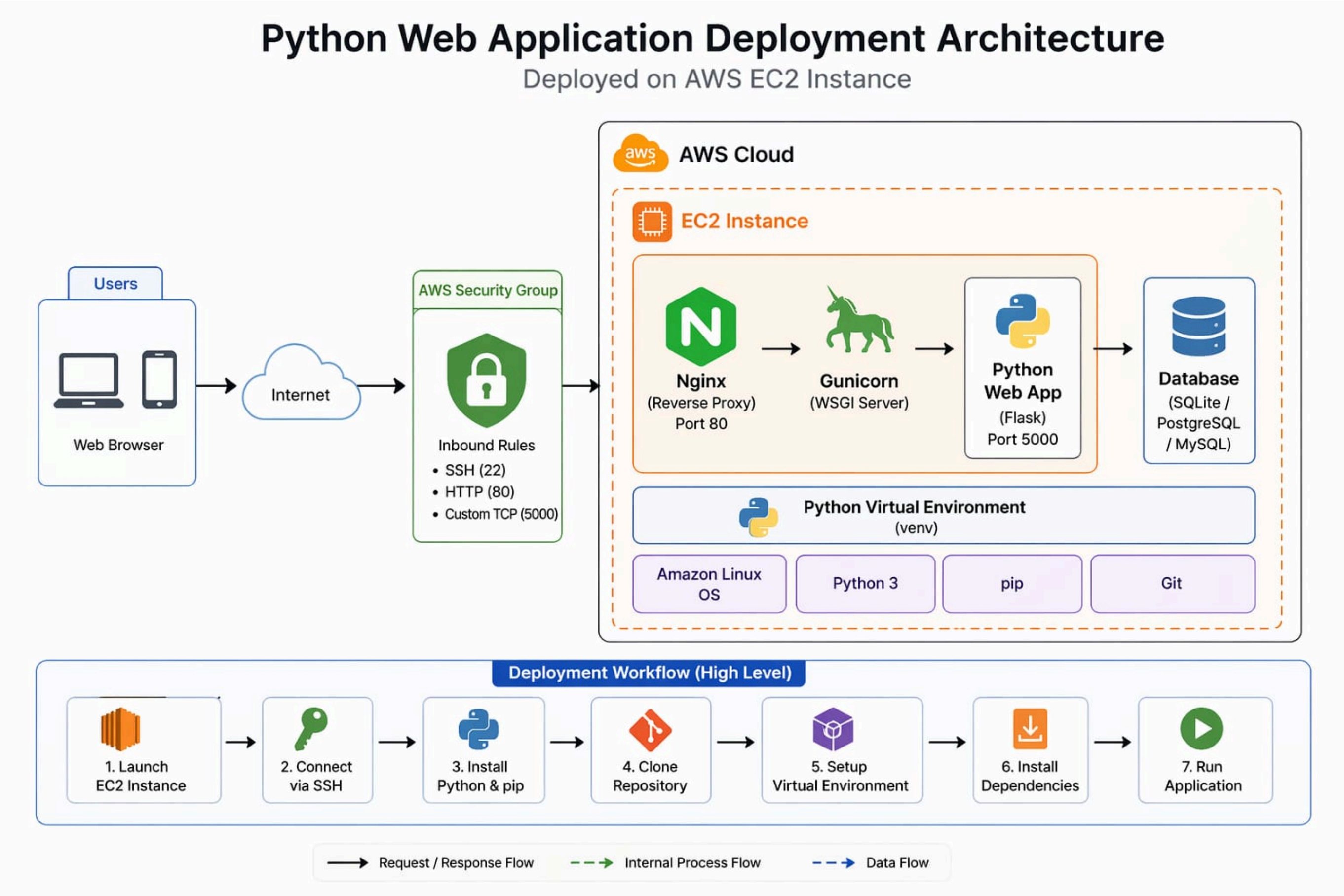
Python Web Application Deployment on Ec2

Introduction

This project demonstrates how to deploy a Python web application on a cloud server using Amazon EC2. It covers the complete process from launching a virtual server to running the application successfully.

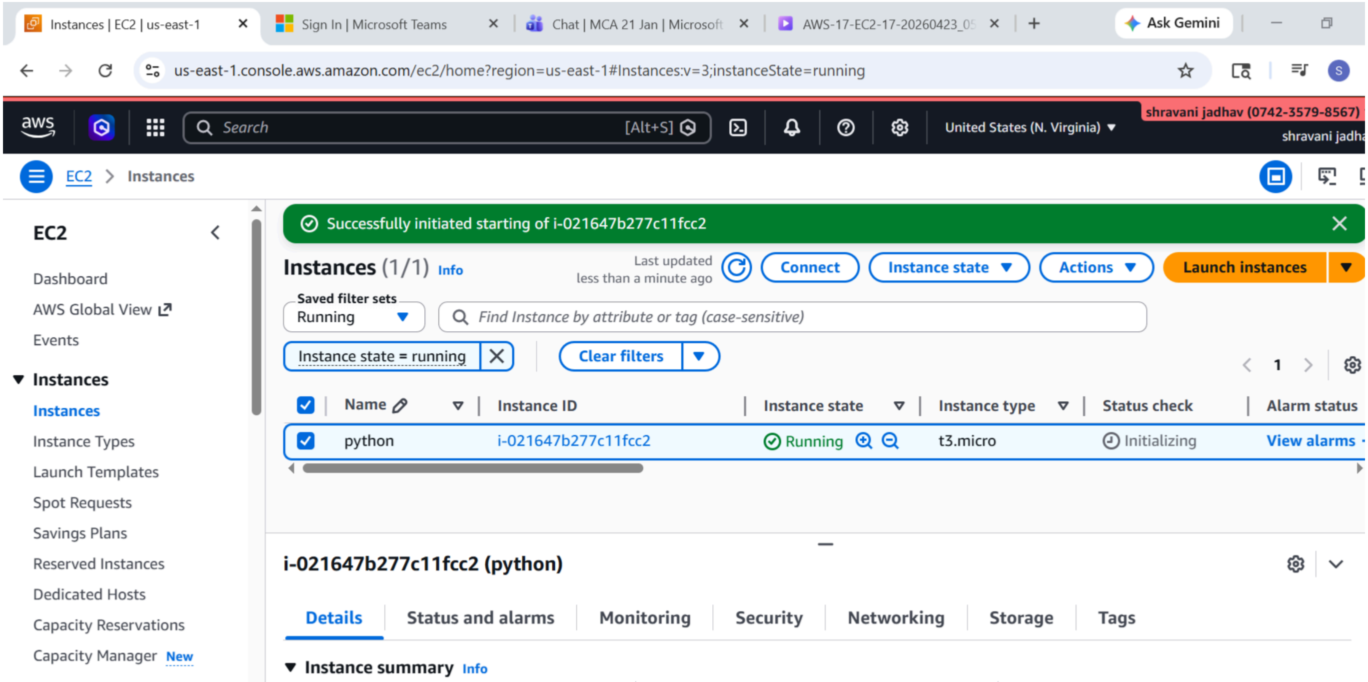
The goal of this project is to gain hands-on experience with cloud computing, Linux commands, and Python environment setup.

Architecture Diagram



Steps I Followed

1.launching an ec2 instance



2.Connect using ssh

ssh -i your-key.pem ec2-user@your-public-ip

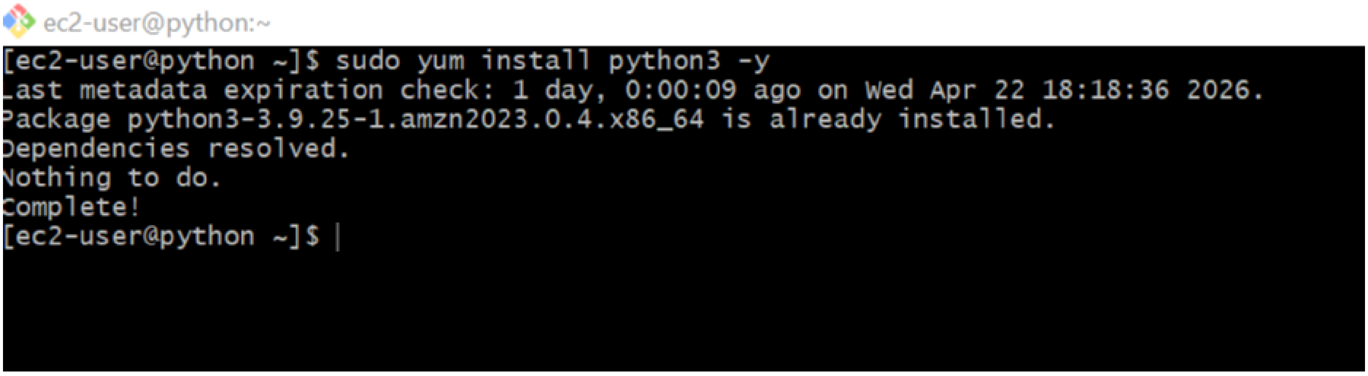
ec2-user@python:~

```
admin@DESKTOP-3UR9NOK MINGW64 /c/shravani_workspace/ssh-key
$ ssh -i "second key pair.pem" ec2-user@ec2-35-172-137-208.compute-1.amazonaws.com
The authenticity of host 'ec2-35-172-137-208.compute-1.amazonaws.com (64:ff9b::23ac:89d0)' can't be established.
ED25519 key fingerprint is: SHA256:yMZJUuUD+eh3Xj3q1P3fFAqdWXNEzaAR0AFQYoEt/Oo
This host key is known by the following other names/addresses:
    ~/.ssh/known_hosts:91: ec2-52-90-106-165.compute-1.amazonaws.com
    ~/.ssh/known_hosts:93: ec2-100-30-183-226.compute-1.amazonaws.com
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added 'ec2-35-172-137-208.compute-1.amazonaws.com' (ED25519) to the list of known hosts.
** WARNING: connection is not using a post-quantum key exchange algorithm.
** This session may be vulnerable to "store now, decrypt later" attacks.
** The server may need to be upgraded. See https://openssh.com/pq.html
   ,     #_
   ~\_  ####_        Amazon Linux 2023
  ~~  \_#####\
  ~~     \###|
  ~~       \#/ ___   https://aws.amazon.com/linux/amazon-linux-2023
   ~~       V~' '->
    ~~~         /
      ~~._.   _/
         _/ _/
       _/m/'
Last login: Thu Apr 23 07:19:47 2026 from 152.58.32.149
[ec2-user@python ~]$ |
```

3.Installation of PYTHON

sudo yum update -y

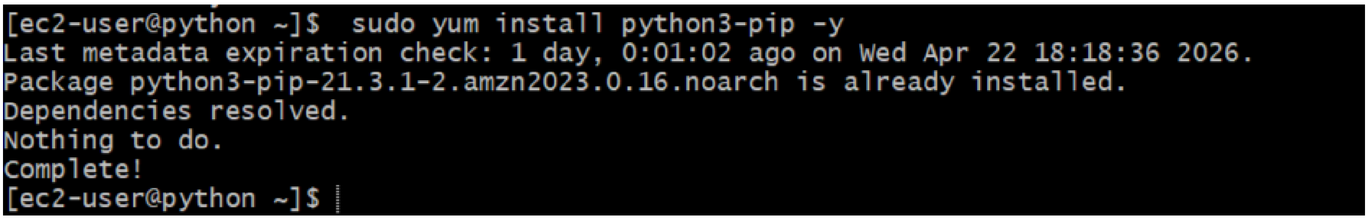
sudo yum install python3 -y



- verify version:
  python3 --version

4.Installation of pip

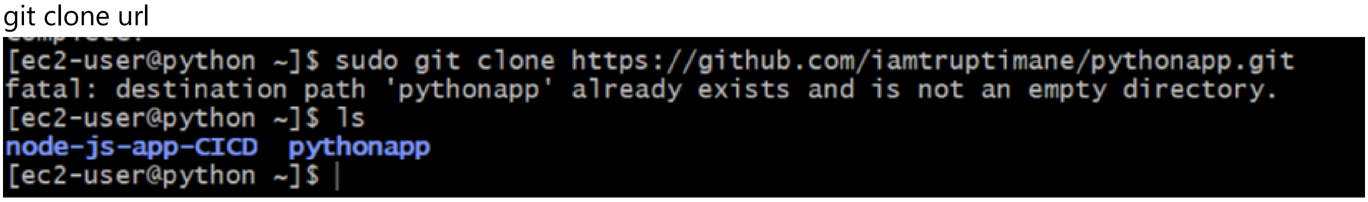
sudo yum install python3-pip -y



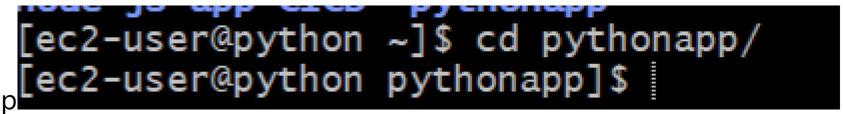
- Verify pip:
  pip3 --version

5.Clone the Repository

git clone url

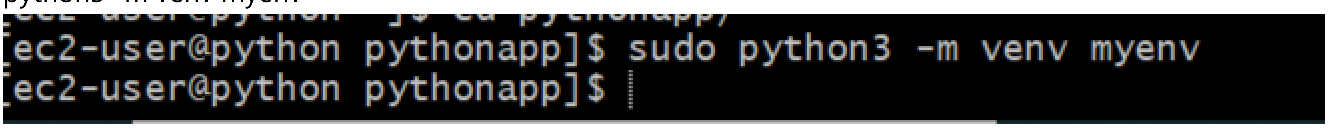


cd pythonapp



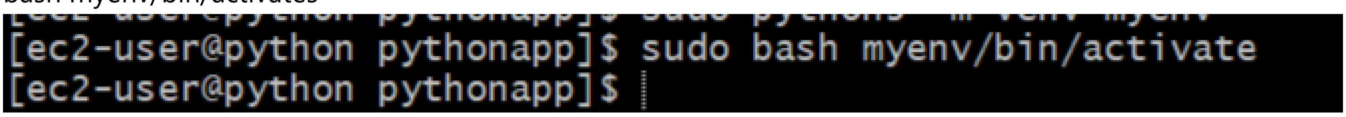
6.Create Virtual Environment

python3 -m venv myenv



7.Activate virtual environment

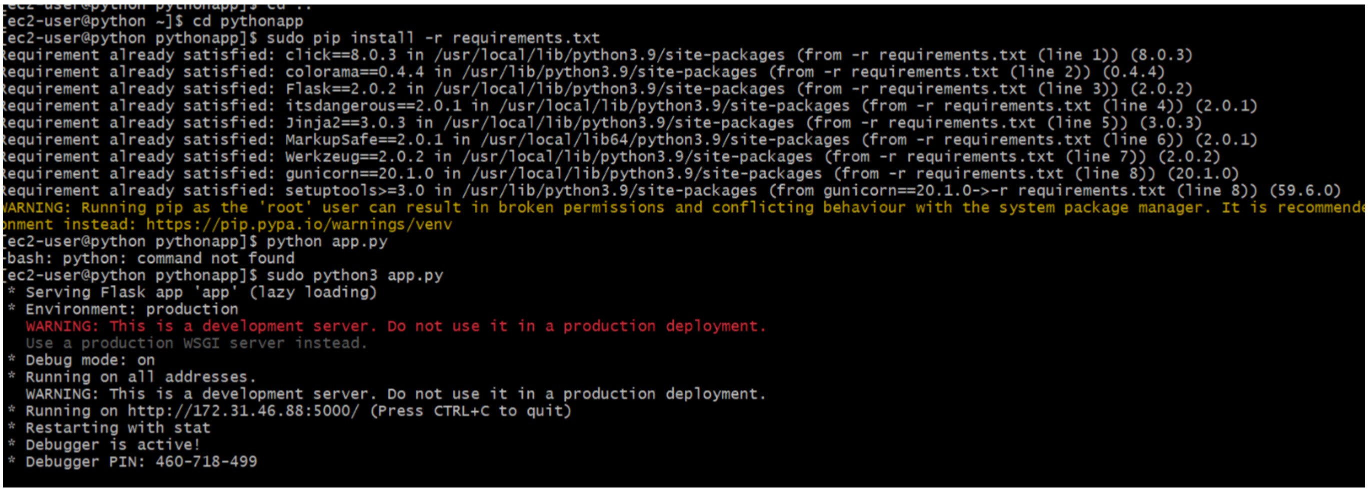
bash myenv/bin/activates



8.Install dependencies

pip install -r requirements.txt

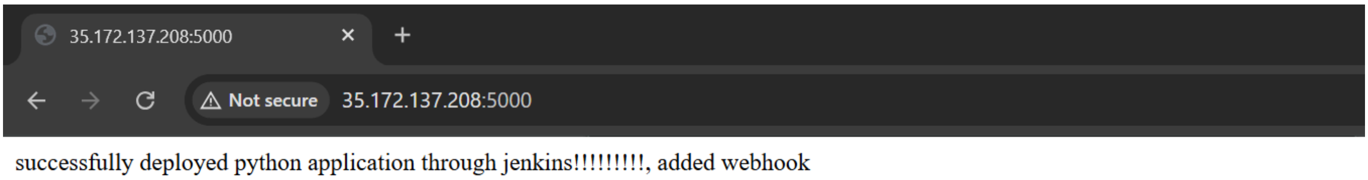
python3 app.py



9.Access the Application

public ip : port

port no -> 5000



Key Features

- Deployment on AWS EC2
- Complete setup from scratch
- Virtual environment usage
- Dependency management with pip

Challenges Faced

- Connecting securely using SSH
- Managing Linux commands
- Setting up Python environment correctly

Project Summary

This project successfully demonstrates the deployment of a Python web application on a cloud platform. Starting from launching an EC2 instance, the process includes installing required tools, configuring the environment, and running the application. It provides practical experience in cloud computing, server management, and backend deployment, making it a strong foundation for real-world DevOps and backend development tasks.